

Seeking Principled Diffeomorphic Hylogisms (6/6): Early Empirical Testing

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The author's approach to this 6-part series is to post each part's technical sketch first without proofs, presuming at each step that prior sections hold. Should later scrutiny invalidate prior sections, the author will diminish or amend claims accordingly. The reader should not treat these early iterations as settled.

Abstract

This research note is the sixth and final in a series that examines the bridge between the physical universe's symmetric informational descriptions. En-route to a diffeomorphic connection from the state configuration space to the SVS and discrete semantic categorisation, this series has proposed a functorial hylogism, allowing the principled identification of physical structures across space and scales. The current note actions that framework, demonstrating the hylogism's explanatory power on a set of empirical tests.

Falsifiable Hypotheses

This series has produced a framework that supports the assertion of several key hypotheses. If reality truly does coexist in a principled and structured parallel with the informational universe, then the paired similarities from I must be observable in P . Recall that a cognitive architecture, like a brain or CPU, is a physical object and that the geometric deviation of its component structure yields an SVS metric warp and probe categorisation. Several hypotheses follow:

Hypothesis – 1 (trivial semantic recovery): *A physically-measured observer object s_{obs} should induce, via probe interactions, a well-defined semantic geometry $g_{sem,obs}$ and categorisation C_{test} .*

Hypothesis – 2 (human label recovery): *A physically-measured observer s_{obs} structurally similar to a human observer s_{human} should have commensurate probe-responses. So, the hylogism-induced semantics via the interaction $\star : s_{obs} \times s_{probe}$ should resemble pre-assigned human similarity measures.*

Hypothesis – 3 (human label recovery via in – sample training): *One may train a structural neural approximator $\hat{s}_{obs}$ (e.g. via computationally constrained renormalisation [1]) to produce human-like semantics via $\hat{g}_{sem,obs}$ and $\hat{C}_{test}$.*

Hypothesis – 4 (human label recovery via out – of – sample prediction): *The trained approximator $\hat{s}_{obs}$ will accurately predict semantic metric differences and categorisations for unseen objects that are deliberately omitted from the training set, but included in the test set.*

Hypothesis – 5 (neural surrogate limit): *In the limit, a sufficiently detailed neural approximator exactly replicates neuro-physiological progressions in a target human brain such that the induced semantic metric $\hat{g}_{sem}^*$ and probe-categorisation C^* exactly predicts accurately-reported human semantics.*

Acknowledgments and Declarations

The author did not use AI to prepare this document. AI has, however, contributed to the development of prior formalisms in this series (see parts 1-5 for further details.)

References

- [1] P. Tohouri, “Computationally Constrained Renormalisation in Virtual Brain Reconstruction,” Mar. 2026. DOI: [10.6084/m9.figshare.31764994.v15](https://figshare.com/articles/preprint/Computationally_Constrained_Renormalisation_in_Virtual_Brain/31764994). [Online]. Available: https://figshare.com/articles/preprint/Computationally_Constrained_Renormalisation_in_Virtual_Brain/31764994.